

Persistent pain after total temporomandibular joint replacement surgery: clinical characteristics, comorbidities, and risk factors.

PMID: 39237445 | Indexed in PubMed

Chronic post-surgical pain (CPSP) after temporomandibular joint (TMJ) surgery is an under-recognized problem. The aim of this study was to document the characteristics of CPSP and identify patient risk factors and comorbidities associated with the development of CPSP after total TMJ replacement (TJR). This was a retrospective cohort study of patients who underwent TJR between 2000 and 2018 at Massachusetts General Hospital, Boston, USA. The primary outcome was the presence of CPSP and use of pain medications after TJR. The secondary outcome was the risk factors associated with the development of CPSP. A total 88 patients were included (79 females, 9 males). The mean follow-up was 4.2 years. Overall, 68 (77.3%) had CPSP and 20 (22.7%) had no CPSP. Of those with CPSP, 32.4% had severe pain and 45.6% continued to take pain medications. Of the 27 patients with data available on the characteristics of the pain, the majority had myofascial pain, while some developed neuropathic pain. A significant difference was noted between the CPSP and non-CPSP groups in terms of preoperative pain, smoking behavior, and use of opioids, non-steroidal anti-inflammatory drugs, muscle relaxants, and neuropathic pain medications.

Temporomandibular Joint Ankylosis Can Be Successfully Treated With TMJ Concepts Patient-Fitted Total Joint Prosthesis and Autogenous Fat Grafts.

PMID: 26878364 | Indexed in PubMed

To measure and identify factors associated with treatment outcomes for patients with temporomandibular joint (TMJ) ankylosis treated with TMJ Concepts patient-fitted total joint prostheses and autogenous fat grafts. This retrospective cohort study evaluated records of patients with TMJ ankylosis from a single private practice, treated from 1992 to 2011, who met the following inclusion criteria: 1) radiographic evidence of bony ankylosis, 2) limited incisal opening, 3) minimum of 12 months' follow-up, and 4) treatment with TMJ Concepts (Ventura, CA)/Techmedica (Camarillo, CA) total joint prostheses and fat grafts. For each patient, the number of previous TMJ surgical procedures, as well as the estimated age of ankylosis onset, age at surgery, and length of postoperative follow-up, was recorded. Subjective evaluations were made with Likert-like scales (from 0 to 10) for 1) TMJ pain, 2) headache and facial pain, 3) jaw function, 4) diet, and 5) disability. Objective evaluations included maximal incisal opening and excursion movements. Nonparametric statistics were used for analysis. There were 32 patients (22 female and 10 male patients) with 48 ankylosed TMJs (16 bilateral and 16 unilateral) in this study, with a mean age of 39 years (range, 11 to 68 years), 2 or more previous TMJ surgical procedures in 69%, and a mean follow-up period of 68 months (range, 12 to 168 months). Trauma was the major etiology of TMJ ankylosis, occurring in 17 of 32 patients (53%). The following improvements occurred: The median value for TMJ pain changed from 8.0 preoperatively to 1.5 at longest follow-up; headache, from 8 to 3.5; facial pain, from 8 to 4; jaw function, from 8 to 2.5; diet, from 7 to 3; and disability, from 7 to 1.5. The median incisal opening was 14.5 mm (interquartile range, 6.3 to 20 mm) preoperatively and 35 mm (interquartile range, 30 to 40 mm) at longest follow-up. The median left lateral excursion improved from 0.5 to 2 mm, and the median right lateral excursion improved from 1 to 1.3 mm. All of these improvements were highly significant ($P < .001$, Wilcoxon tests). Equally favorable outcomes were found in patients with 12 to 48 months of maximal follow-up and patients with more than 48 months of maximal follow-up. The treatment of TMJ ankylosis with the TMJ Concepts patient-fitted

total joint prosthesis in combination with fat grafting around the articulation area of the prosthesis is a viable and predictable method for improving pain levels, function, and quality of life, as well as prevention of reankylosis of the TMJ.

Total Alloplastic Temporomandibular Joint Replacement.

PMID: 34776679 | Indexed in PubMed

End-stage temporomandibular joint (TMJ) disease are not uncommon and affects quality of life. Multiple surgical procedures have been mentioned in literature for management of TMJ disease which ranges from conservative management to aggressive resection of involved joint and replacement with alloplastic total joint prosthesis. The purpose of the present paper was to provide an overview of the role of alloplastic total joint prosthesis in TMJ replacement. Alloplastic total joint prosthesis is nowadays considered as a standard of care in the adult patients who require TMJ replacement. The requirement of alloplastic total prosthesis has increased in present era with the improvement in design and material of implants, surgical skills and reported victorious outcome along with improved quality of life after its use. It provides restoration of form and functions, improvement in quality of life, reduction in pain and maintenance of ramal height. Additionally, in TMJ ankylosis it reduces chances of re-ankylosis and allows facial asymmetry correction. Currently, enough evidence is however not available for replacement in skeletally immature patient. The authors conclude that the total joint replacement is a standard procedure for end-stage TMJ disease. Every maxillofacial surgeon should be well-acquainted with TMJ replacement.

Temporomandibular joint replacement: a New Zealand perspective.

PMID: 24332584 | Indexed in PubMed

Alloplastic total temporomandibular joint replacement (TMJ TJR) has been performed in New Zealand utilizing the TMJ Concepts patient-fitted system since 2000. The data analysed in this study were collected retrospectively from questionnaires sent to all maxillofacial surgeons in New Zealand who had implanted TMJ Concepts devices between 2000 and 2011. A total of 63 devices were implanted in 42 patients (13 males, 29 females) during this 12-year period. The primary indication for TMJ TJR was end-stage joint disease resulting from ankylosis and arthritis. The mean age of the patients was 47 years (range 7-80 years). The most common complication reported was transient facial nerve impairment in 4.8% of the patients. Objective results, measured as the maximal incisional opening, improved by a mean of 17.3mm ($P<0.01$); 90% of patients reported improved quality of life. New Zealand oral and maxillofacial surgeons have concluded that TMJ TJR using the TMJ Concepts prosthesis is a reliable treatment option for the management of end-stage TMJ disease.

[Total temporomandibular joint replacement with alloplastic prosthesis].

PMID: 23234748 | Indexed in PubMed

Three total temporomandibular joint (TMJ) replacement surgeries were performed on two patients with TMJ ankylosis using W. Lorenz system. The first surgery was performed on a female patient with osseous

ankylosis who had no mandibular movement since she was 18 months old. The second surgery was on a patient with fibrous ankylosis who had an interincisive opening distance of 12 mm. Her condition was the result of a three year old subcondylar fracture which didn't receive any treatment. The first patient had total replacement of both TMJs, while the other just one TMJ. The follow up until today has been of eight months with an interincisive opening of 29 mm in the first patient and 34 mm in the other. There has not been any mandibular deviation during opening, nor postoperative pain in both cases. Facial nerve affection was not permanent in both patients. For all of this, we can conclude that total TMJ replacement is an effective treatment with stable results in patients with TMJ ankylosis.

Temporomandibular joint reconstruction with total alloplastic joint replacement.

PMID: 21332746 | Indexed in PubMed

This paper is a preliminary paper which presents the early findings of an ongoing prospective trial on the use of the TMJ Concepts and Biomet Lorenz total joint replacement systems for the reconstruction of the temporomandibular joint (TMJ). Total alloplastic replacement of the TMJ has become a viable option for many people who suffer from TMJ disease where surgical reconstruction is indicated. Degenerative joint diseases such as osteoarthritis, rheumatoid arthritis, psoriatic arthritis, TMJ ankylosis, malunited condylar fractures and tumours can be successfully treated using this technique. There are a number of TMJ prostheses available. Two of the joint replacement products, which have been found to be most reliable and have FDA approval in the United States, are the TMJ Concepts system and the Biomet Lorenz system, and for this reason they are being investigated in this study. This study presents the findings of seven patients with a total of 12 joint replacements using either the TMJ Concepts system or the Biomet Lorenz joint system. Two patients (3 joints) had the TMJ Concepts system and five patients (9 joints) had the Biomet Lorenz system. Although still early, the results were generally pleasing, with the longest replacement having been in position for three years and the most recent six months. The average postoperative mouth opening was 29.7 mm (range 25-35 mm) with an average pain score of 1.7 (range 0-3, minimum score of 0 and maximum 10). Complications were minimal and related to sensory disturbance to the lip in one patient and joint dislocation in two patients.

[Temporomandibular joint prostheses].

PMID: 33739661 | Indexed in PubMed

Temporomandibular joint prosthesis. The temporomandibular joint can be affected by end-stage pathology such as arthritic disease, trauma, ankylosis. The main debilitating symptoms results in limited mouth opening and pain. TMJ prosthesis is recommended for disease resulting in an anatomical modification and aims at restoring joint function with pain relief. Despite its initial history of health scandals, prosthetic replacement for end-stage TMJ disease is gradually becoming a common procedure because of good functional results and low morbidity. Significant advances have been made in the design of the temporomandibular joint prostheses during the last decades, specially thanks to the use of 3D printing. The indications should be further clarified and extended.

Total temporomandibular joint replacement in recurrent temporomandibular joint ankylosis: a case report.

PMID: 37496634 | Indexed in PubMed

Total temporomandibular joint replacement (TMJR) is a surgical procedure in which end-stage temporomandibular joint disorders are replaced with an alloplastic prosthesis between the mandible and the base of the skull when autogenous grafts are inadvisable. These alloplastic prostheses may be available as stock or custom-made prostheses consisting of the mandibular condyle and glenoid fossa components. Although the first total temporomandibular joint prosthesis was used in the 1960s, we present the case of a 20-year-old female patient, probably the first case of total temporomandibular joint arthroplasty in Nepal, performed at Dhulikhel Hospital in Kavrepalanchok. The patient underwent bilateral TMJR with a custom joint prosthesis for recurrent TMJ ankylosis. Postoperatively, the patient noticed significant improvements in mouth opening, chewing ability, facial esthetics and, most importantly, her self-esteem and confidence.

Custom made/patient specific alloplastic total temporomandibular joint replacement in immature patient: a case report and short review of literature.

PMID: 36591882 | Indexed in PubMed

Temporomandibular joint reconstructive surgery in a growing patient represents a challenging situation. Autogenous and alloplastic reconstructive options are being studied in literature; however, there are still some limitations. The objective of this case report is to evaluate a novel custom-made prosthetic system in a 12-year-old TMJ ankylosis patient. The patient had complaints of temporomandibular joint ankylosis and hypoplasia. The patient had already been operated two times with autogenous grafts. Swelling and tumefaction were apparent on the right side of the face. Mouth opening was 1.5 centimeters, with limitations in lateral and protrusive movements of the jaws. Hypertonic muscles and pain upon palpation were registered. There were no signs of luxation, fracture, or traumatic avulsion. After examination, unilateral TMJ ankylosis was apparent on TC scans. Revision surgery was planned with the use of true plastic temporomandibular joint customized prosthesis. The patient underwent a TMJ reconstruction surgery using CAD/CAM custom-made patient specific prosthesis. The follow up period of this patient was 46 months and showed successful healing with no complications. Replacement of TMJ with custom made alloplastic material that is reported can be considered as a safe and useful option for growing young individuals in selected cases.

Long-Term Stability in Temporomandibular Joint Replacement: A Review of Related Variables.

PMID: 39590422 | Indexed in PubMed

Background: The temporomandibular joint (TMJ) is a key component of the stomatognathic system, playing a major role in maintaining mandibular stability and function. Temporomandibular disorders (TMDs) are a prevalent disease in the world, with surgical treatment being reserved for complex cases or end-stage TMJ disease. A narrative review was conducted to describe the quantitative and qualitative factors that affect TMJ prosthesis stability. Methods: Studies with a sample size equal to or greater than

10 subjects who underwent surgical procedures for joint replacement using stock or customized ATM prostheses were included. This narrative review examined some variables that may influence in terms of the longevity of the TMJ prosthesis, highlighting issues to be considered in future research. Results: The current development of TMJ prostheses is benefiting from technological advances, offering a suitable adaptation to the patient's anatomy and superior results in functionality and patient satisfaction. However, the biomechanical complexity of the TMJ shows unique challenges compared to other joints in the body, where anatomical, biomechanical, and functional requirements are high. The stability of the TMJ prosthesis is affected by multiple variables, including the selection of biocompatible materials that resist corrosion and wear, the design of the prosthesis, the diagnosis and indication for its use, and the surgeon's experience. The success of TMJ replacement can be measured by improving the patient's quality of life, reducing pain, restoring mandibular functionality, and recovering suitable facial morphology for the patient's conditions. Conclusion: There is a need for training of maxillofacial surgeons in TMJ surgery and replacement, as well as a greater focus on the research and development of systems to simplify surgical design and procedures and to optimize the results of TMJ replacement.
